# BABU BANARASI DAS UNIVERSITY

# LUCKNOW

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SCHOOL OF COMPUTER APPLICATION

# PROJECT
# ON

## APPLY END TO END CLOUD SECURITY TO APPLICATONS

## (BCACSN14201)


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# CASE STUDY

## Data Protection and cloud security

1. A healthcare company stores sensitive data of patients in cloud. Analyse how data anonymisation, tokenisation, DLP can be used to protect data. Identity possible risk and solution.

- Annoymization: Irreversible strips identifiers (name / DOBs). Best for research / analytics where individual identity isn't needed.
- Tokenization: Replaces sensitive data (SSNS) with a "token". Best for internal operations where systems need to link records without seeing the raw data.
- DLP (Data Loss Prevention): scans for medical codes or private info in real-time. Best for preventing accidental leaks or unauthorized sharing.

| Risk | Solution |
|---|---|
| 1. Accidental leaks | DLP policies to block unauthorized emails / uploads. |
| 2. Lateral Movement | Zero trust access and strict encryption at rest |

② A cloud Based organization faces frequent network attacks, design a secure architecture using SDN, Vxlan, IDS

Ans ① SDN (software defined Networking)

SDN separates the control plane & data plane, allowing centralized management of the network.

benefits :-

- easy traffic monitoring
- Fast response to attack.
- Dynamic security policy update
- Better : visibility of network traffic.

eg - If suspicious traffic is detected, SDN controller can immediately block it.

② Vxlan (virtual extensible LAN) :-

Vxlan creates, secure, virtual network segments across cloud infrastructure.

Benefits

- network isolation b/w departments.
- secure communication b/w virtual machine.
- Reduce lateral movement of attackers.

eg - finance, HR & Admin systems can be separated using vxlan.

(3) IDS (Intrusion detection System)

IDS monitors network traffic & detects suspicious activities.

Types :

- Network IDS
- Host IDS

Benefits

- detects malware
- Identifies unauthorized access.
- Alert administrators in real-time.

eg - Detecting repeated failed login attempts or unusual traffic spikes.